

ML Lab Week 14: CNN Image Classification

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SECTION: F

COURSE: MACHINE LEARNING

DATE: 20/11/2025

1. Introduction

The objective of this lab was to design, train, and evaluate a Convolutional Neural Network (CNN) capable of classifying images of hand gestures into the categories rock, paper, and scissors. Using the Rock–Paper–Scissors dataset provided through Kaggle, the task involved completing the given PyTorch boilerplate, preprocessing the dataset, building the CNN architecture, training the model, and analyzing its performance on unseen test data.

2. Model Architecture

The CNN model consists of three convolutional blocks followed by a fully connected classifier.

Convolutional Layers

Each block includes:

A Conv2D layer : Kernel size: 3×3 Padding: 1 (to preserve spatial dimensions)

ReLU activation

MaxPool2D with kernel size 2×2, which reduces the spatial dimension by half at each block

The channel progression across the convolutional layers is $3 \rightarrow 16 \rightarrow 32 \rightarrow 64$

The input images are resized to 128×128, and after three pooling operations the feature map size becomes 16×16, producing a flattened feature vector of size : $64 \times 16 \times 16 = 16384$

Fully Connected Classifier

The classifier block contains a flatten layer, a Linear layer reducing the feature vector from 16384 $\rightarrow$ 256 , ReLU activation ,Dropout ($p = 0.3$) to prevent overfitting ,Final Linear layer (256 $\rightarrow$ 3) corresponding to the three gesture classes.

This architecture creates a balanced and efficient CNN capable of extracting visual features and performing classification.

3. Training and Performance

- Training Hyperparameters
- Optimizer: Adam
- Loss Function: CrossEntropyLoss
- Learning Rate: 0.001
- Batch Size: 32
- Number of Epochs: 10

During training, the loss steadily decreased across epochs, showing consistent learning and convergence.

Final Results

- Test Accuracy: 97.26%

This high accuracy indicates that the model learned the gesture representations effectively and generalized well to unseen data.

4. Conclusion and Analysis

The CNN model performed exceptionally well, achieving a 97.26% test accuracy, demonstrating strong feature extraction and classification capability for the Rock–Paper–Scissors dataset. The training process was smooth, and the model converged quickly with stable loss reduction.

Challenges Faced:

Ensuring images were properly normalized and transformed before training.

Potential Improvements

1. Data Augmentation: Applying transformations like rotation, flipping, and random cropping could improve robustness.
2. Batch Normalization: Adding BatchNorm layers in each convolutional block may stabilize training and improve accuracy.
3. More Training Epochs: Increasing the number of epochs may allow the model to learn more complex patterns.